Keep the equation. Slow the distraction.

Technical report - 22 October 2026
Joseph Ayanda - original code and authored lesson, with AI assistance



The task. A creator has a flattened lesson clip and no editable timeline. A repeated decoration competes with the explanation. A whole-clip slowdown also changes the graph, counter and narration. Frame Review proposes a spatially limited 5x retime, asks the creator to mark what must stay unchanged, then tests the encoded comparison against the original source.

The demonstration. An authored 12-second lesson graphs y = -2x + 1. Its graph marker and step counter change on their original timeline. A separate corner repeats every six frames. The source has disclosed neural narration; it is synthetic educational media, not a real teacher trial.

Live demo: https://le-d72c965de52749b08fd72a2513bccce1.ecs.us-east-1.on.aws
Judge-accessible source: https://github.com/josepha-mayo/opencv-frame-review

From a proposed region to a measured file

Analyze	Review	Approve	Verify
OpenCV 5 proposes a region and interval.	Name up to four fixed protected rectangles. Inspect a static overlay and readable plan.	Bind explicit consent to source, prediction and recipe hashes.	Encode, decode again, check preservation, display original/control/edited crops.

Substantive OpenCV work. OpenCV handles luminance transformations and masked regional measurements, candidate geometry and visual overlays. The verifier uses image differences, masked histograms and PNG encoding/decoding for the actual exported boundary crops. FFmpeg performs controlled decoding, encoding, probing and audio copying. NumPy represents native planes and exact deterministic mappings. This is a deterministic vision workflow; it does not claim a trained semantic model or an autonomous agent.

A bounded retime. The local periodic mode requires at least three repeated cycles, period at most 30 frames, per-plane recurrence mean error at most 2 native codes, maximum individual recurrence difference at most 32 codes, and sufficient luminance range. It repeats one observed cycle 5x more slowly within the detected interval. The output mapping and every outside native sample are checked independently. The recurrence budgets can still admit small nonperiodic changes.

Preservation is a gate. Declared rectangles require zero decoded RGB error against the original source across every frame. The unchanged control is checked separately. Supported AAC-LC mono/48 kHz audio is copied, with stream metadata, packet payload/timing/skip metadata and decoded PCM comparisons. A failing retiming-preservation gate withholds the edited download.

Human review. Selecting a source or checking a plan never approves an edit. Correcting an area invalidates consent and the old plan. A protected rectangle overlapping the edit is refused. Reload recovery reopens a job through reads, without repeating upload or approval. Unsaved rectangles and consent clear. Browser testing caught and corrected a strict-CSP overlay geometry defect that DOM stubs had missed.

Full-frame preservation does not imply readability, semantic equivalence, clinical safety or publishing clearance. Entry and exit transitions remain visible for review. No such certification is claimed.

Evaluation that can be reproduced

Case	Audio	Result
320x180 / 1.0 seconds	Silent	Exact native mapping / outside samples
320x180 / 4.1 seconds	AAC tone	Exact mapping, packet and decoded-audio checks
320x180 / 12.0 seconds	AAC tone	Exact mapping, packet and decoded-audio checks
640x360 / 2.4 seconds	Silent	Exact native mapping / outside samples
640x360 / 9.0 and 12.0 seconds	AAC tone	Both duration cases passed
Authored math lesson / 12.0 seconds	Neural narration	All 360 frames of all four named areas exact

Nine verification stages passed. Backend units; six duration exports; duration refusals; original audio; original silent media; audio refusals; pixel refusals; the authored lesson; and studio UI/HTTP checks. The pinned Linux image ran 137 backend unit tests and 56 studio tests, with one Windows-specific test skipped. Windows ran all 57 studio tests. Several UI checks are shared with the historical suite; these totals are not independent experiments.

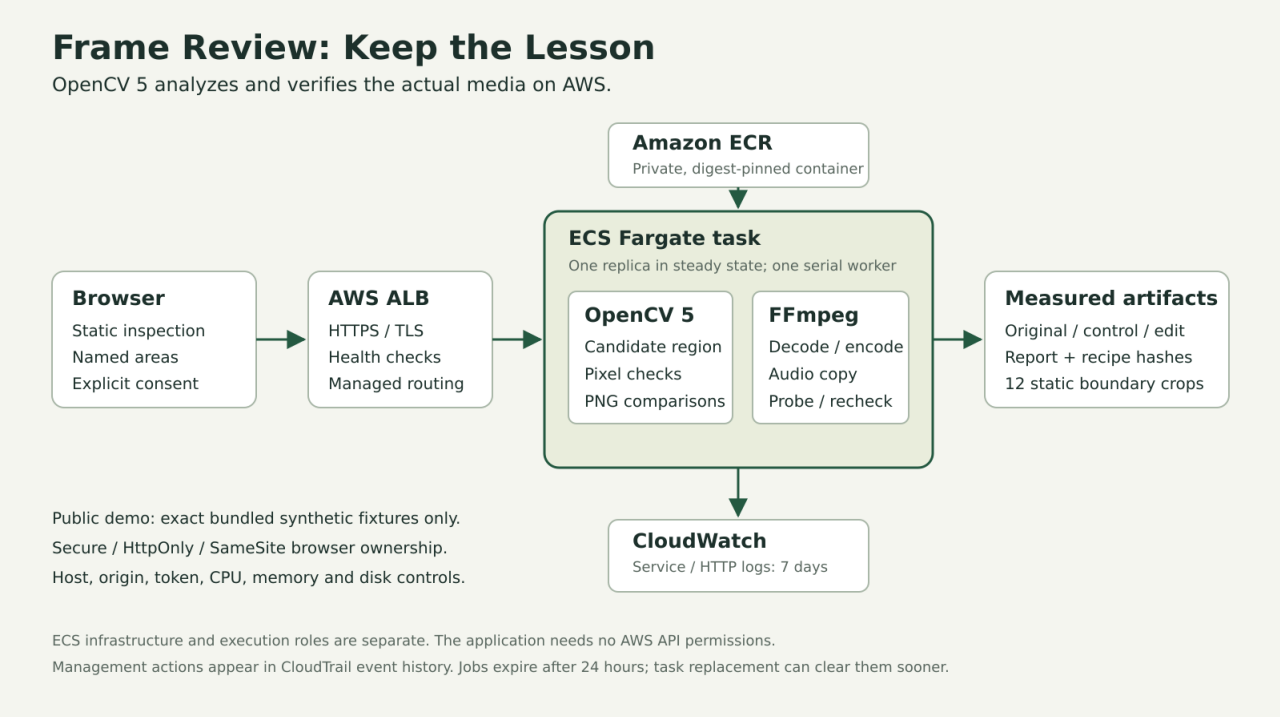
Math-lesson measurements. Observed cycle: 6 frames; retimed cycle: 30. The selected rolling trigger count changes from 10 to 2 under the frozen development metric. All declared equation, explanation, graph and counter RGB errors are zero across 360 frames. Every native outside sample, the intended mapping, audio packet/decoded-sample data and video timestamps passed independent comparisons. All twelve PNG response hashes matched. Adjacent unprotected RGB samples had maximum code error 2; this is disclosed chroma interpolation, not hidden as exact global RGB preservation.

Controlled whole-clip baseline. A 5x slowdown makes the 12-second source 60 seconds long and moves its first counter change from 3 to 15 seconds. The local edit remains 12 seconds and keeps that change at 3 seconds. The compared counter patches match the source bytes. The baseline script and receipt are included; this is one authored-fixture experiment, not a manual-editor or user study. An initial baseline encode produced 1799 instead of 1800 frames; an explicit tail pad and frame-rate filter corrected that separate offline baseline.

Retained negative evidence. Verification on a Windows bind mount hit the unchanged worker wall-time limit on a 12-second case. On native Linux storage, the same source and limits passed. Historical failed recipes and the historical missing screenshot were not relabeled as success. The active build has its own explicit source manifest.

Browser and AWS receipts are separate from unit/media checks. The live AWS lesson completed through the real browser UI; its downloaded output matched both its report and the independent local lesson export. No creator time-savings or learning-outcome result has been measured.

AWS delivery, reproduction and limits



Actual cloud execution. ECS runs the OpenCV/FFmpeg analysis and export, rather than serving only a static page. One Fargate replica runs in steady state; managed canary deployments can temporarily run two. Execution and infrastructure roles are separate. The application needs no AWS API permissions. The dedicated infrastructure policy supplement scopes ECS updates to this service and PassRole to its dedicated roles.

Bounded public demonstration. Before starting a media worker, the public endpoint accepts only byte-exact bundled synthetic fixture hashes. A Secure, HttpOnly, SameSite=Strict session cookie owns each job; a different browser cannot read a known job ID. Existing host/origin/request-token checks remain enforced. One serial worker, eight jobs, disk caps and bounded request accounting limit load. Inactive jobs expire after 24 hours; task replacement can clear them sooner. HTTP/service logs have seven-day CloudWatch retention. ECS management actions are visible in CloudTrail event history.

Reproduce. Build the supplied Dockerfile. It pins Python 3.12.14, OpenCV 5.0.0.93 and NumPy 2.3.5, and verifies the exact Debian FFmpeg/ffprobe hashes. Run `python studio/run.py --verify /tmp/qualification` in the image on native Linux storage. The source archive includes tests, fixtures, goldens, generation provenance and compact receipts. The live web demo needs no AWS account or Docker setup.

Limits. Local admission remains 320x180 or 640x360, 30 fps, 1-12 seconds in 0.1-second steps, canonical limited-range BT.709 H.264/YUV420 with optional narrow AAC metadata. Many ordinary clips are refused. The frozen multimedia stack has known package-scan findings; it is not exposed as an arbitrary public media processor. Fixed regions do not track moving content. Real teacher trials, competent manual-edit baselines, broader intake and broader playback support are the next evaluation, not completed claims.

Source manifest: d00b0c7753e4bd8debcee5ba2186c06d32530333ba1d9320376c41faaf529966
Container image index: sha256:64cc5f02e99464140539bec0fde686b808fcc27b1146939d13abf18c07415f5a
Lesson output: b6e8052ec70f60e2189f41645167111fb846e88092727bf91213e7621fcda68e